



How to give a good presentation?

Interpretable Machine Learning Group
Chair for Statistical Learning and Data Science
LMU München

Some parts of this presentation are adapted from earlier presentations prepared by Thomas Brox, Frank Hutter, Andre Biedenkapp, Katharina Eggensperger and Matthias Feurer.



Why should you aim for a good presentation?



→ You'll have to give a lot of presentations in your life (in academia **and** industry)

Such presentations can decide whether

- You get a job
- You get a promotion
- Your favourite project gets funded
- You get the resources you need
- ...
- You get a good grade ;-)



Photo by [Matthew Jungling](#) on [Unsplash](#)

What matters is what your audience gets (not which points you “covered”)

- Often, the audience is not as interested in the topic as you
- You’ll have to tell them why they should care
- If nobody cares or understands **it’s typically your own fault**

At least the key points must get across to everyone
Some details may only be for experts, that’s OK

1. Structure is key
2. Adapt your talk to your audience
3. Present in pictures
4. Readable slides
5. Practice, Practice, Practice!
6. Check your technical equipment before
7. Behave naturally
8. Learn from the mistakes of others

High level to low level to high level

- Catch your audience's attention
- Then tell them what you'll tell them and why they should care (priming)
- Then tell it to them
- Then tell them what you just told them

Make transitions clear, don't forget the "meta-talk"

- E.g., *In order to explain X, first I'll need to explain Y*
- E.g., *Now that we've seen X and Y, we have the ingredients to do Z*
- Remind the audience where you are in the talk, e.g. using a reoccurring outline slide
- Use meaningful titles

Don't get lost in details

- In case of doubt leave out some details
- Use a "T-structure": combine broad coverage of a topic with depth about one (or few) key aspects
- Focus on what you find most interesting

Start your presentation with

- a brief introduction of yourself
- a motivation of why your topic matters and why the audience should care
- what you will talk about (outline slide only after the motivation)

End your presentation with

- the main takeaways
- an outlook
- a clear statement announcing the end of your presentation,
e.g. *That's it from my side and now I am happy to answer questions*
- a Thank-you slide is not necessary, better show the conclusion/discussion slide
(unless you thank collaborators)

The paper you are presenting is written for a specialized research community.

Your audience often has a different background

- “Customize” the motivation (and ideally connect it to the topic of event/prior talk)
- Cover the necessary background
- We are experts on some topics – don’t bore us with what we already know

In general

- A talk to the CEO is completely different than one to the tech support group
- A talk applying method X to problem Y is completely different when you’re talking to community studying X or Y

Slides full of text are hard to follow

- The audience will read and **not listen to you**
- Reduce text, use **more images**
- Use animation to guide focus of attention

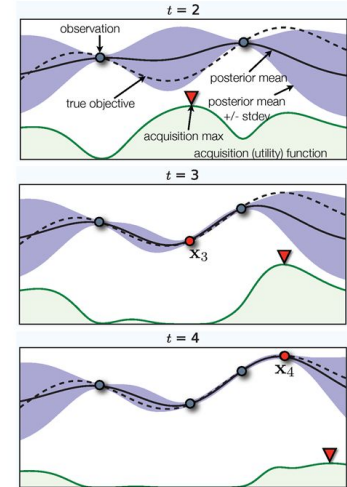
Method of Choice: Bayesian Optimization

- Prominent approach to optimize expensive blackbox functions [Mockus et al., '78]
- Approach
 - Observe a few function evaluations
 - Construct a probabilistic model of the objective function, for example a Gaussian process
 - Use that model to compute a so-called acquisition function that quantifies how useful a new data point is, trading off exploitation of areas predicted to be good and exploration of areas where the model is uncertain
 - Use the acquisition function to select the next point to evaluate the function at
 - Evaluate the function there, refit the model, and iterate
- Efficient in the number of function evaluations
- Works when objective is nonconvex, noisy, has unknown derivatives, etc
- Recent convergence results [Srinivas et al, '10; Bull '11; de Freitas, Smola, Zoghi, '12]

15

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14

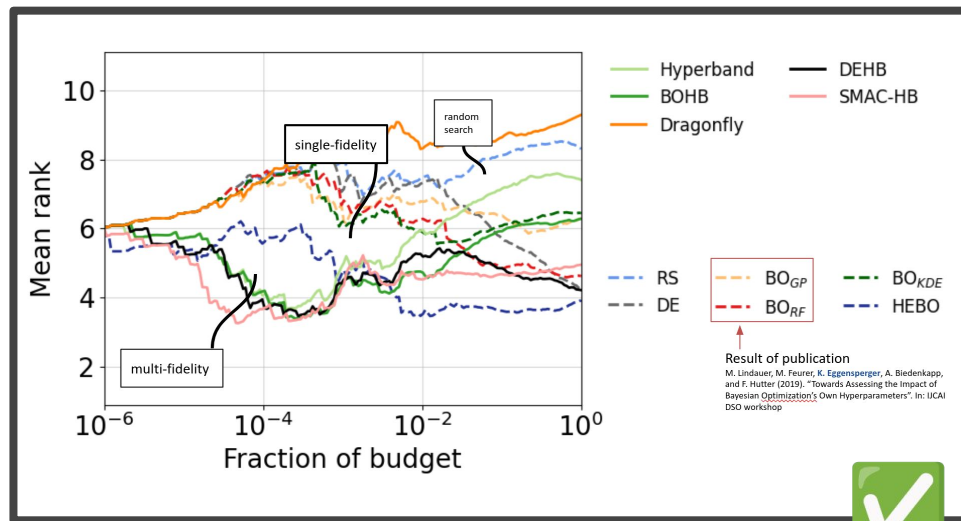
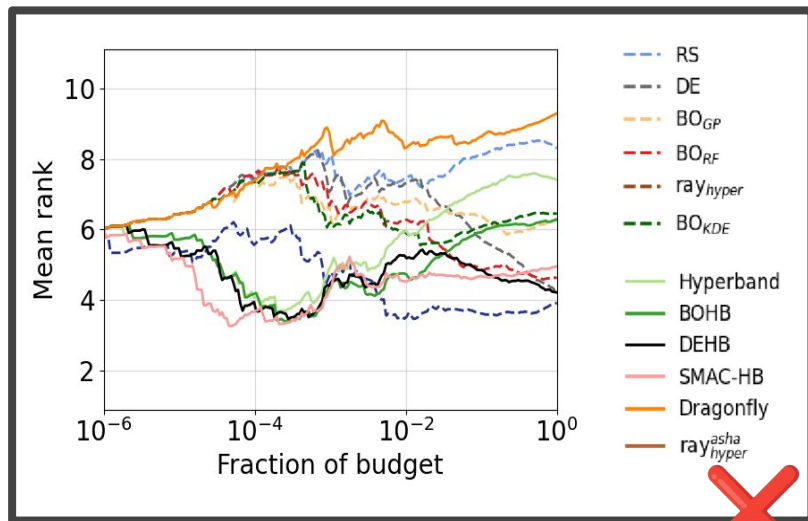
#3 Present in Pictures



How to present a graph/plot?

→ The same applies to equations and tables

- always explain what the graph shows (ideally with some visual highlighting)
- use presenter to guide audience
- take enough time



PD VS. ALE

PD:

$$f_{S,PD}(x_S) = \mathbb{E}_{\mathbf{x}_{-S}} \left(\hat{f}(x_S, \mathbf{x}_{-S}) \right)$$

ALE:

$$f_{S,ALE}(x) = \int_{z_0}^x \mathbb{E}_{\mathbf{x}_{-S} | x_S = z_S} \left(\frac{\partial \hat{f}(z_S, \mathbf{x}_{-S})}{\partial z_S} \right) dz - \int \tilde{f}_{S,ALE}(x_S) d\mathbb{P}(x_S)$$

- Recall: PD directly averages predictions over marginal distribution of $\mathbf{x}_{-S}$
- ALE is faster: Needs $O(2 \cdot n)$ model calls vs. $O(n \cdot g)$ for PD with g grid points
- Difference 1: ALE averages the
 - prediction changes (via partial derivatives, estimated by finite differences)
 - over conditional distribution $\mathbb{P}(\mathbf{x}_{-S} | x_S = z_S)$
- Difference 2: ALE integrates these partial derivatives over $z_S \in [z_0, x] \subseteq \mathcal{X}_S$
 $\rightsquigarrow$ isolates effect of x_S and removes main effect of other dependent features
- Difference 3: ALE is centered so that $\mathbb{E}_{x_S} (f_{S,ALE}(x)) = 0$

Feature Effects - ALE plots



#3 Present in Pictures, and use animations



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Text, Font & Color

1. Make sure to pick a reasonable text and background color
2. Can you read this? Also from the back?
Remember: Contrast & resolution of your laptop is better than that of the projector (so test it before)
3. Sometimes the font size is too tiny. This also applies to axis labels.
4. Pick a single font type and color and use it throughout *the whole presentation*
5. Highlight important **keywords** when there is a lot of text, but: choose a consistent way of highlighting

Graphics

1. Size up figures to use most of the slide.
2. Add some bullet points so its self-contained.
3. Not all animations are useful.
4. Screenshots are okay, if you do not have access to the original image.

Other

1. Make sure there are no typos in your slides
2. A list needs more than one entry
 - e.g. this is not a list!
3. Make sure slides are self-contained (important for most presentation types)
4. Have slide numbers
5. Have only little content per slide

1. Plan each part!

- Have a time budget
- Have bullet points with the main points
- Practice & check the timing for the part

2. Put it all together and practice!

- Do the transitions work?
- Always get stuck at the same point? Change that point!
- Don't speak too fast! Speaking too slowly is almost impossible
- Make use of breaks

3. Finetune start and beginning!

- Know how you want to start
(when you're most nervous)
- Know how you want to end
(what the audience remembers)

★ Bonus tips

- Starting at a random slide of your presentation
- Stand and use presentation mode (as realistic as possible)
- Think about potential questions
- Have backup slides with left out details

Checklist

- Do you have to bring your own laptop?
 - Does your laptop work with the projector?
 - Do you have the right dongle?
 - Instant messaging (Teams, Zoom, Slack, Mattermost, etc) switched off?
 - Desktop free of too personal items?
 - Screen saver switched off?
 - Enough battery or laptop plugged in?
 - Are all updates installed?
- Is your presentation in the right format?
- Do all videos show properly?
- Does audio work?
- (if applicable) Does your laser pointer work?

★ Bonus tips

- Prepare and test your equipment before the talk!
- Have your slides also as a PDF ready



Keep **eye contact** with the audience; don't turn your back

→ But do not wonder what they might think of your presentation! (now it's too late)

Relax!

Answering questions:

- Listen to the whole question carefully; don't interrupt
- Repeat what you understood, especially for long/multiple questions.
- Think before you answer
- Short and precise
- If you don't know the answer, say so. This is okay.

 **Bonus tipp:** Ask someone to take a video of you presenting and watch it

Have you ever been to a presentation where you were

🤔, 😞, 🙄 or irritated by a specific behaviour of the presenter?

Then

- Analyze what went wrong
- (if possible) give them (friendly & constructive) feedback
- Do not make the same mistakes

★ **Bonus tipp**

If you see a great presentation, learn from it
(and let the presenter know that you enjoyed the presentation)

1. Start with **something positive**

- In your own reviews you don't want to hear only negative things, either
- People are more receptive to criticism after hearing something positive

2. Make **concrete suggestions**

- Bad example: *The lecture was bad*
- Good example:

I couldn't follow the math because I couldn't read your handwriting on the board – better use a projector or slides



Mistakes, that are easy to avoid:

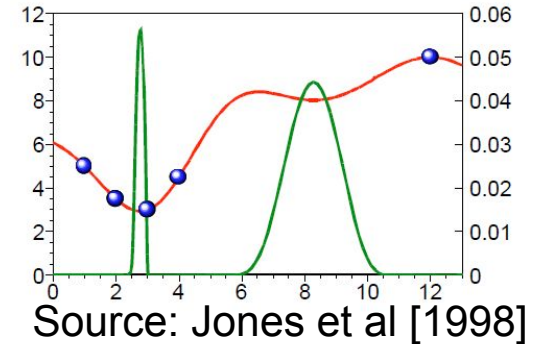
- Talking too slow/fast/loud/quietly
- Bad time management, e.g. a 15 minute talk should last 15 minutes
- Preparing too many slides, e.g. 100 slides for 15 minutes
- Not spending enough time on a slide
- Technical difficulties
- Improvising

1. Never present other people's work as your own

- Never copy-paste (even critical if it is your own work)
- State explicitly what is your contribution

2. Give appropriate credit

- references for figures
- licence for photos/icons
- Quotes: *X and Y [12] define this problem as follows : "..."*



→ Never cheat or plagiarize on purpose, clearly mark your references, adopt best practices for avoiding mistakes

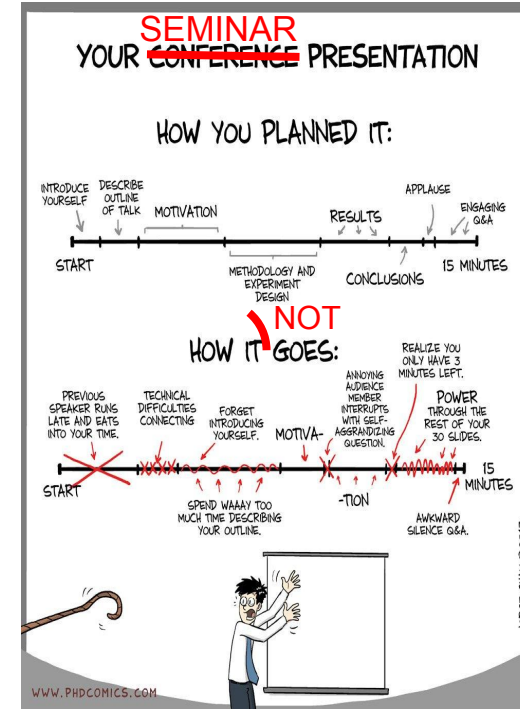
During the presentation

- First & Last 30s sec of your presentation
- How well did you motivate your problem
- How well did you explain technical details

During Q&A

- It is part of your presentation.
- Good things to do:
 - Think about questions that you might want to discuss with the audience? Open problems?
 - Prepare answers for obvious questions

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- **LMU Schreibzentrum Workshop**
<https://www.schreibzentrum.fak13.uni-muenchen.de/freitagsworkshops/index.html>
- **How to give a great scientific talk** <https://www.nature.com/articles/d41586-018-07780-5>
- **Free Images** <https://www.pexels.com/>, <https://unsplash.com/>
- **How to read a research paper** <http://ccr.sigcomm.org/online/files/p83-keshavA.pdf>